
[CS3704] Software Engineering

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Virginia Tech
9/17/2026

Announcements

- **PM1.2.1 due Thursday (9/24) at 9:30am**
- **Midterm Exam (10/1) on Canvas**
 - Timed 1-hour exam (start before 10:59pm!)
- **PM1.2.2 due Monday (10/5) at 11:59pm**
- **HW2 due Wednesday (10/7) at 11:59pm**
- **More Ut Prosim opportunities posted**

Class Updates: Dr. Brown Travel

Tues, 9/22: Project Workday

Thurs, 9/24: Project 1 Demos

Tues, 9/29: Exam Review [No Class, dcbrown travel]

Tues, 10/1: Exam [No Class, dcbrown travel]

- No instructor office hours during 9/28-10/2, but please feel free message me on Mattermost or email with any questions.

Project Update: Lightning Talks

How to create an effective pitch?

A pitch deck is a presentation that provides a brief but informative overview of your business (project). It should cover the key points of your business (project) plan, the products and services you (propose to) provide, ~~high-level financial projections, and funding needs.~~ [LivePlan]

(Modified, for this class) The Only 10 Slides You Need in a Pitch

1. Title (project name plus team name and names of team members)
2. Problem/Opportunity
3. Value Proposition (how does your project alleviate the problem)
4. Underlying Magic (novelty compared to existing methods)
5. Competitive Analysis (requirements elicitation techniques and results)

Project Updates: Lightning Talk Rubric

Project	Points
Presenters spoke clearly	5
Presentation slides are readable	5
Team name and members are clearly presented	5
Presenters clearly describe the problem their project tries to solve	15
Presenters explain how their solution addresses the problem	15
Presenters explain the motivation of their approach from requirements elicitation results	10
Presentation mentions related work (similar tools, papers, articles, etc.)	5
Visual representation of the project (at least one use case of a real-world scenario for your system, live or recorded; fully implemented or prototype; etc.)	25
All team members contribute to the presentation	5
Presentation is between 6:00 and 7:00 minutes long	10
	100

9/24 in class!

Project Update: Lightning Talk (Misc.)

- Presentation slides due 9/24 *before class!*
- All team members must present.
- Order will be selected randomly right before class.
- Please try to arrive on time 
- Please stick to the designated time 

Project Updates: Proposal Rubric

Team	Points
Proposal is clearly written and organized	5
Spelling, grammar, etc.	5
Relevant title, abstract and authors (team members) are clearly presented	5
Proposal is in the correct paper format	5
Motivating example providing a scenario of this project and its usage	7
Introduction explains the problem and motivates proposed solution	7
Related work section discusses existing related tools and/or research	7
Proposal provides details on the requirements elicitation results and how they motivated the project	10
Proposal provides details on the preliminary usability study to examine/improve system design	10
Proposal has a section describing the high-level design, implementation, and testing approach	10
Proposal discusses the use of AI (of if AI was not used, why the team decided against it)	10
Discussion section outlining limitations and future work to complete if there were more time	10
Conclusion summarizes the overall report	4
References are included	5
	100

Due 10/5

UI Design

User Interfaces

UI Design Processes

Design Heuristics

Learning Outcomes

By the end of the course, students should be able to:

- **Understand software engineering processes, methods, and tools used in the software development life cycle (SDLC)**
- Use techniques and processes to create and analyze requirements for an application
- **Use techniques and processes to design a software system**
- Identify processes, methods, and tools related to phases of the SDLC
- Explain the differences between software engineering processes
- Discuss research questions and current topics related to software engineering
- Create and communicate about the requirements and design of a software application

Warm-Up

Which software application do you find has the most user-friendly interface, and why?

What about the least user-friendly interface?

Goal: decide the structure of the software and the hardware configurations that support it.

- The *how* of the project
- How individual classes and software components work together in the software system.
 - **User interfaces**, *structural elements*, and *data*
- *Software Artifacts*: wireframes, prototypes, storyboards, task flow analysis, etc.

Review: High-Level Design

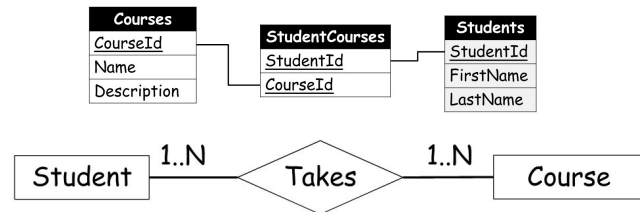
- **Goal:** Design the overall shape and structure of the program to *reduce risks* and *meet requirements*.
- Common program structures:
 1. Pipe and Filter
 2. Event-based
 3. Layered
 - a. Client-Server
 - b. Data-Centric
 - c. Model-View-Controller (MVC)
 - d. Microservices

Review: Database Design

A relational database is a collection of tables.

- Each table contains rows and columns, with a unique key for each row
- Each entity described in a database has its own table
- Each row represents an instance of the entity
- Each column represents an attribute
- Entity-Relationship (ER) Diagrams model database designs

Employee Details			
emp_id	first_name	last_name	address
1	John	Newman	New York
2	Michael	Clarke	Melbourne
3	Steve	Smith	Los Angeles



Review: Low-Level Design

Goal: To decompose subsystems into modules

- Object-Oriented vs Functional Design Principles
 - Class diagrams
- Design Patterns
 - Working solutions to identify classes and instances, their roles and collaborations, and responsibilities in a particular context.
 - Pattern Families:
 - i. Creational
 - ii. Structural
 - iii. Behavioral

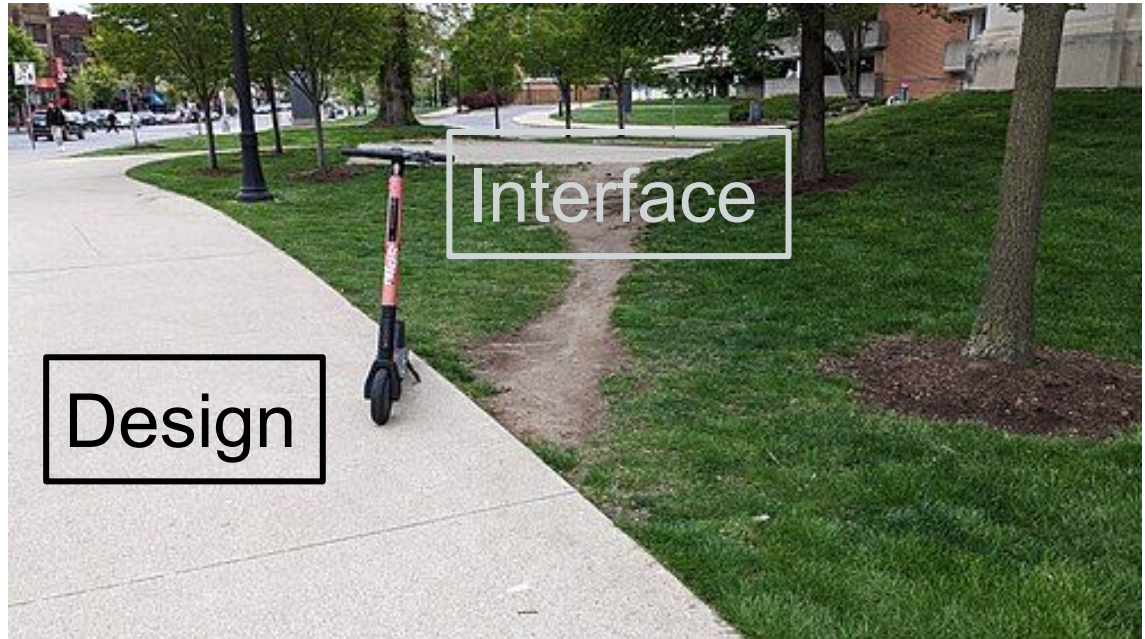
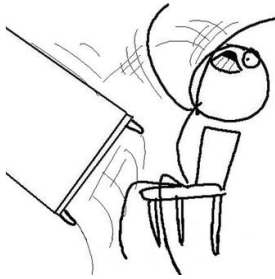
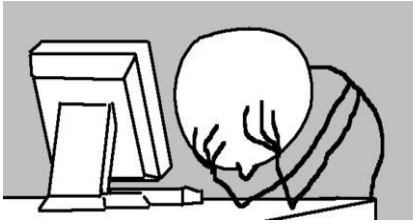
User Interfaces (UIs)

Software engineering is about meeting user needs.

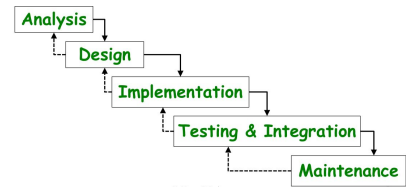
- UIs define the way users interact with the system
 - *All software has a user interface!*

“Design is about leveraging principles about humans (psychology) to build interactions people can use” [Coblenz]

Why Focus on UIs?



Plan-Driven UI Design



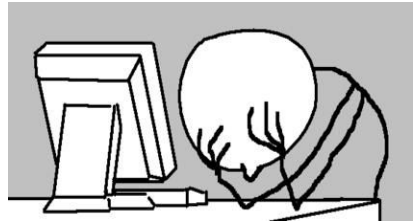
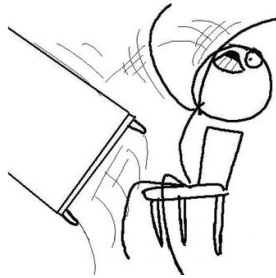
- Waterfall processes are generally poor for user interface design!
 - Plan-driven processes: rigid, comprehensive, slow
 - UI design is *risky*
 - Often will not get it right the first time
 - Users are often not involved
 - No feedback until the end
 - UI flaws cause requirements changes!



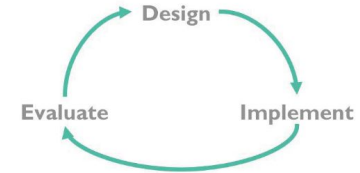
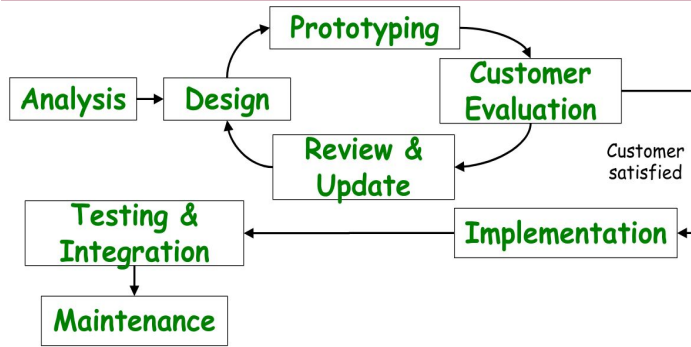
Iterative UI Design

Iterative Design the Wrong Way:

- Feedback (complaints) from users inform the next version design for the next iteration.



Iterative UI Design (cont.)



- **Dual Track:** UI design runs 1–2 sprints ahead of development.
 - Early prototypes can detect usability/UI issues
 - Often with dedicated UI/UX teams
- **User-Centered Design:** Early focus on users and tasks
- More iterations → Better UI
 - Constant evaluation and iterative improvement

Universal Design

- A school of thought that seeks to design for *all* users, as much as possible
 - As opposed to designing for the *typical* user.
- Should not involve “dumbing down” interfaces, but make design better for everyone.
- Guidelines for Universal/Accessible Design
 - Section 508 rules: <https://www.section508.gov/>
 - World Wide Web Content (WC3) Accessibility Initiative: <https://www.w3.org/TR/WAI-WEBCONTENT/>

UI Design Processes

DISCOVER

- Bodystorming
- Cognitive walkthrough
- Contextual inquiry
- Design studio
- Dot voting
- Heuristic analysis
- KJ method
- Metrics definition
- Stakeholder and user interviews

DECIDE

- Affinity diagramming
- Comparative analysis
- Content audit
- Design principles
- Journey mapping
- Mental modeling
- **Personas***
- Site mapping
- **Storyboarding***
- Style tiles
- **Task flow analysis***
- User scenarios

MAKE

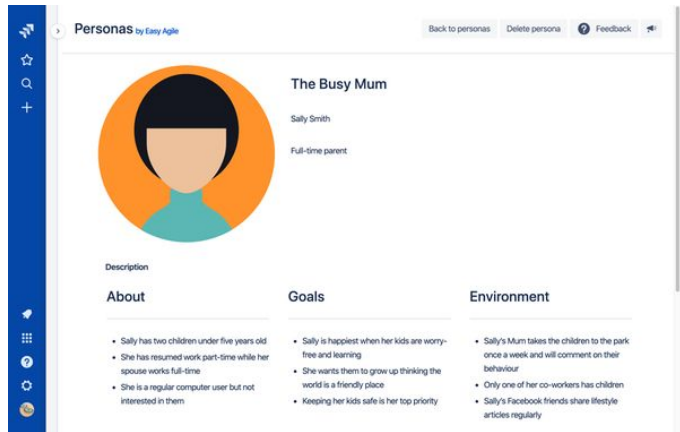
- Design pattern library
- **Prototyping***
- **Wireframing***

VALIDATE

- Card sorting
- Multivariate testing
- Usability testing
- Visual preference testing

Decide: Personas

A **persona** is arch-user type which represents a segment of a user population, and allows role-play during task planning and UX design.



Decide: Storyboarding

A **storyboard** illustrates the *timeline* of user performing a task as a sequence of frames.

Zen: Multiple Courses Storyboard



Sit & Order

First time customers Ann and Andy are taken to their table where they browse the menu on the iPad and place an order for drinks and appetizers.



Beverages Delivered

The waiter brings over their drinks promptly.



Apps Delivered

The appetizers are brought to their table.



Second

Next up entrees.



Dinner is Served

The waiter brings out the entrees.



Swipe & Done

Andy swipes his card to pay and opts for an email receipt.

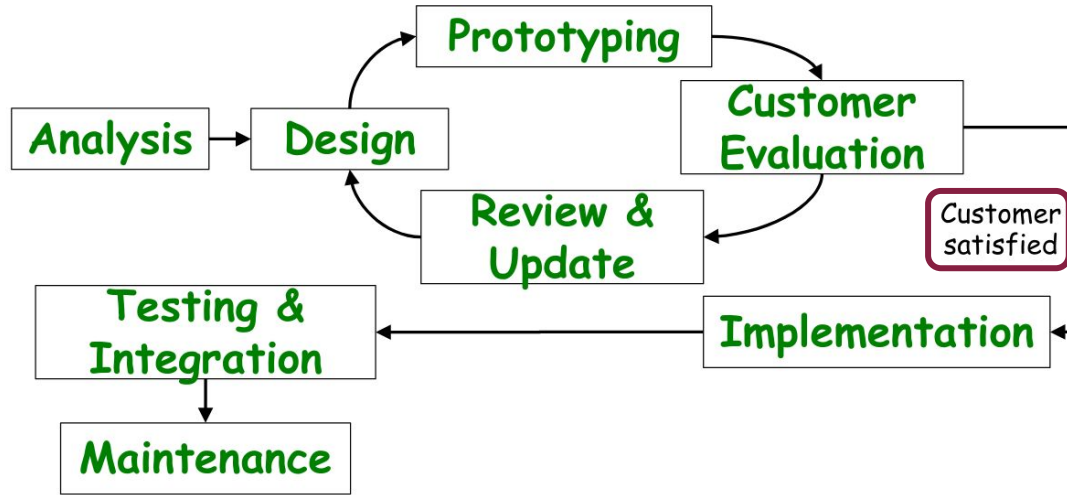
Useful tools:

- Storyboarder
- Plot
- FrameForge
- Miro
- Canva
- StoryboardThat
- ...

A flow map describes the *wayfinding* activity of a user and transitions between UI states.



Make: Prototyping



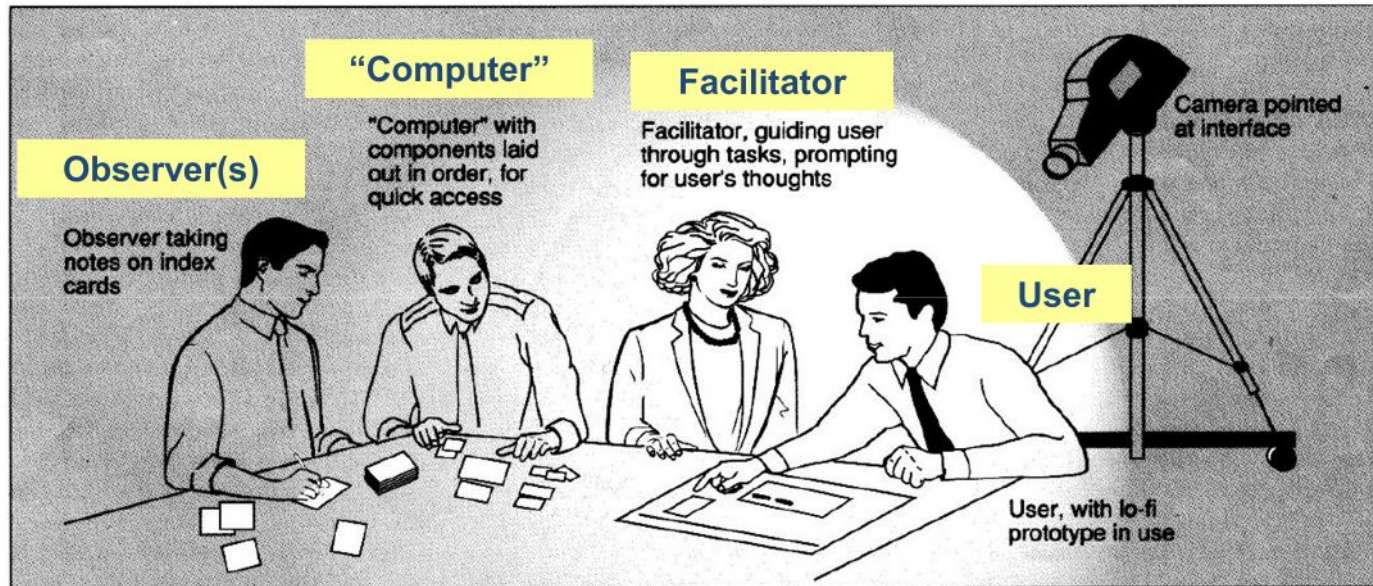
Useful tools:

- Figma
- Miro
- App Builder
- Adobe XD
- Balsamic
- Usability Hub
- ...

Prototype involves clients, design quickly and iteratively before implementation.

Make: Prototyping (cont.)

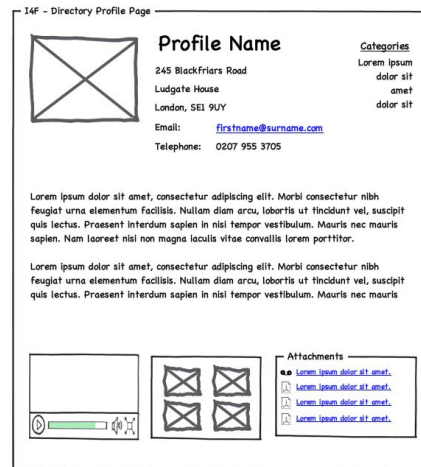
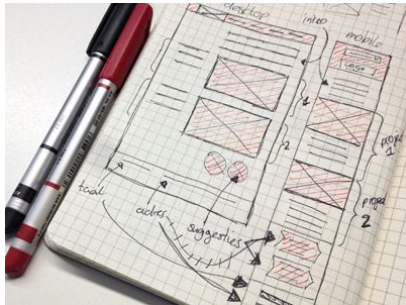
- In the past, companies adopted a process known as “paper prototyping”.



Make: Wireframes

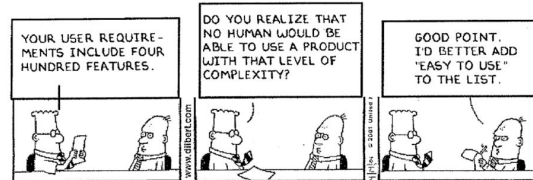
A **wireframe** is a view schematic that captures all layout and content decisions of that view.

- How will you allocate space for particular content?
- Where will content live?
- What actions can you do?



Usability

- **Usability:** a quality attribute that measures how easy a user interface is to learn and use
- Dimensions of usability
 - Learnability: is it easy to learn?
 - Efficiency: once learned, is it fast to use?
 - Visibility: is the state of the system visible?
 - Errors: are errors few and recoverable?
 - Satisfaction: is it enjoyable to use?
 - ...



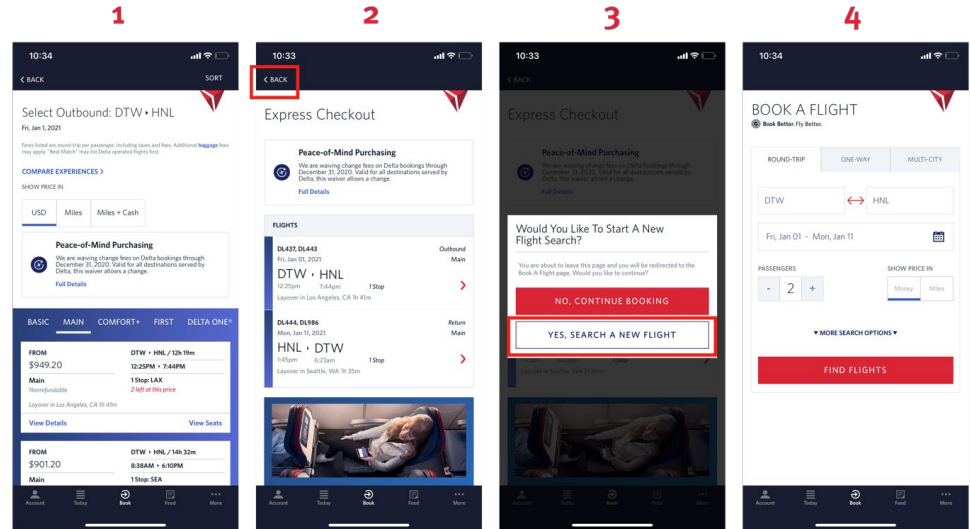
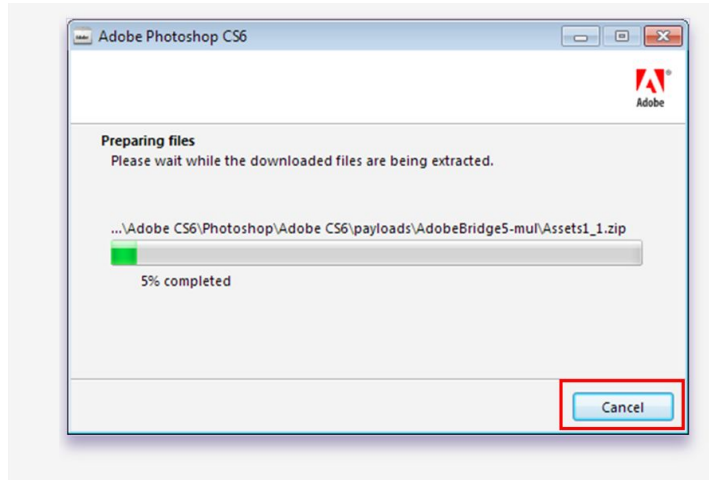
Ten Usability Heuristics

1. Visibility of system status
2. Match between system and the real world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Help users recognize, diagnose, and recover from errors
10. Help and documentation

[Nielsen]

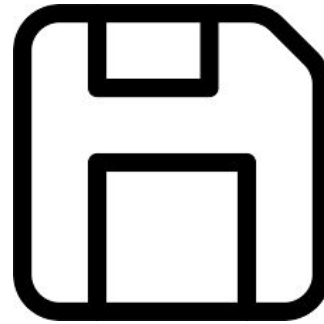
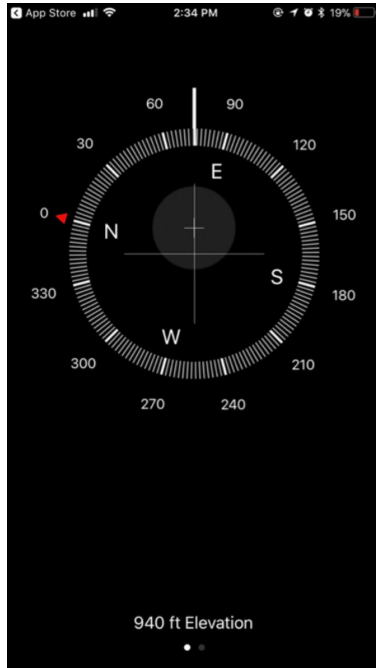
1. Visibility of System Status

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.



2. Match Between System and Real World

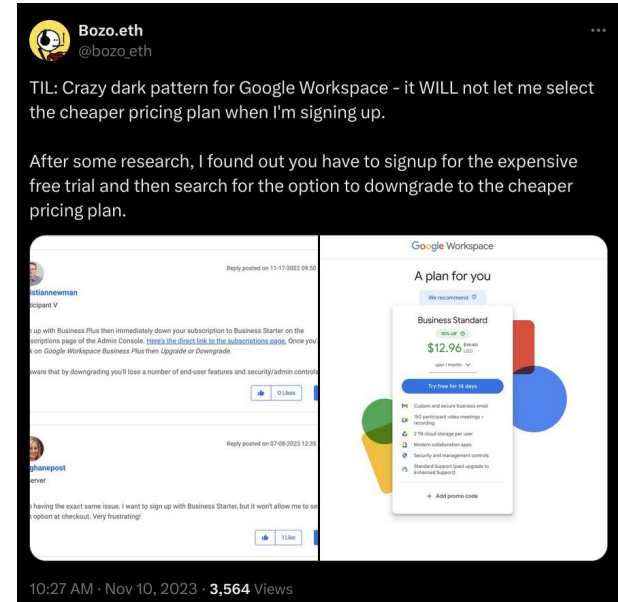
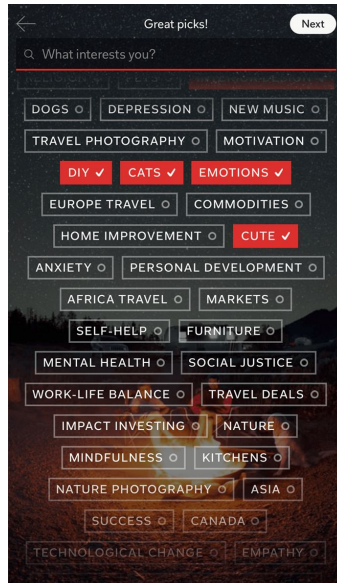
The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order.



Commonly used for save functionality, however floppy disks have been obsolete since 2001! 

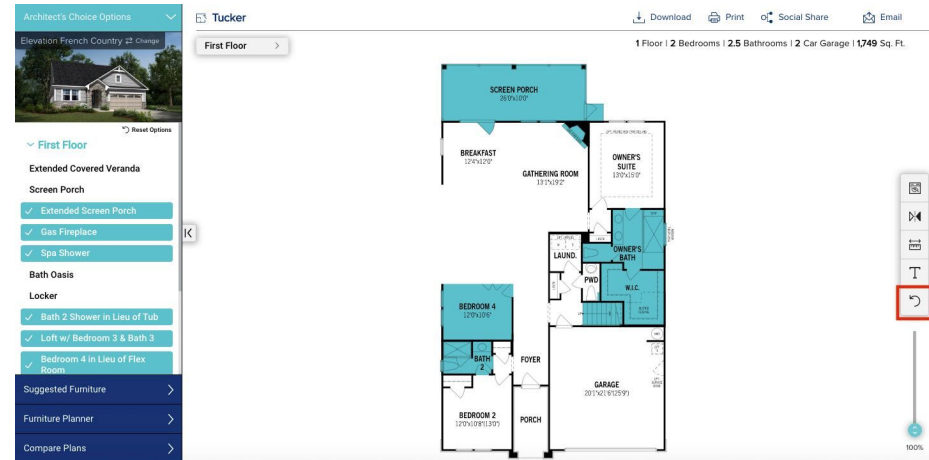
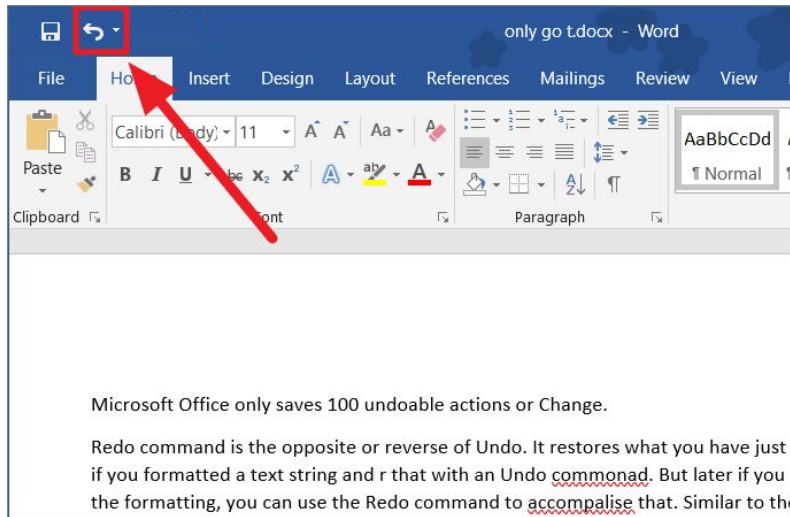
3. User Control and Freedom

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process.



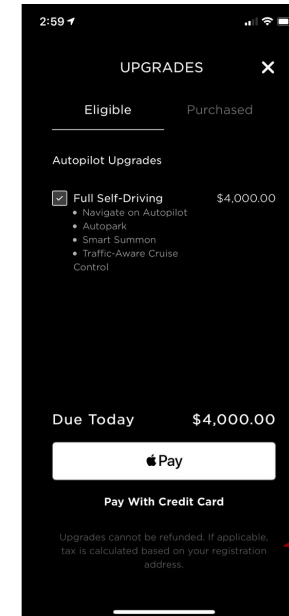
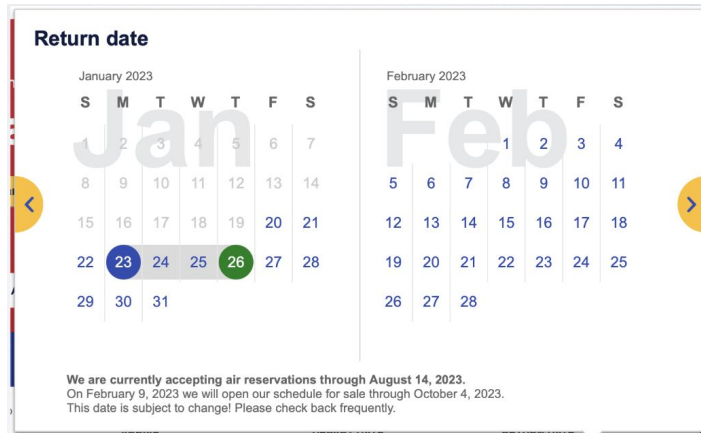
4. Consistency and Standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions.



5. Error Prevention

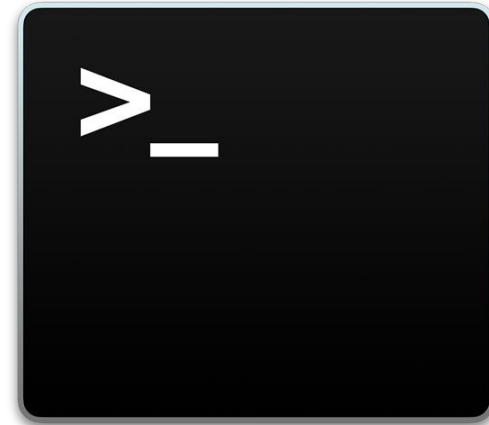
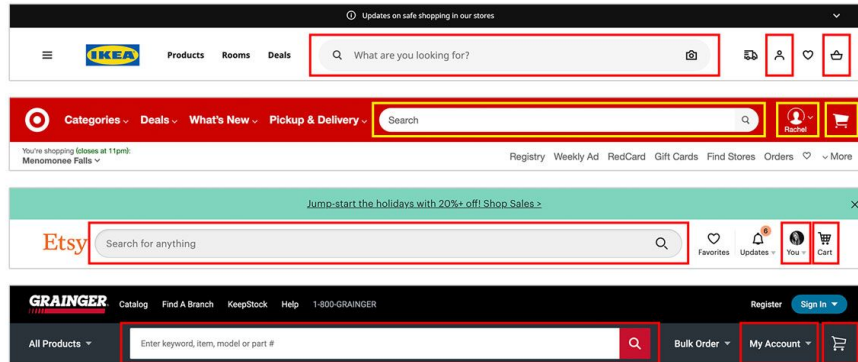
Good error messages are important, but the best designs carefully prevent problems from occurring in the first place. Either eliminate error-prone conditions, or check for them and present users with a confirmation option before they commit to the action.



“Upgrades cannot be refunded...”

6. Recognition Rather than Recall

Minimize the user's memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another. Information required to use the design (e.g. field labels or menu items) should be visible or easily retrievable when needed.

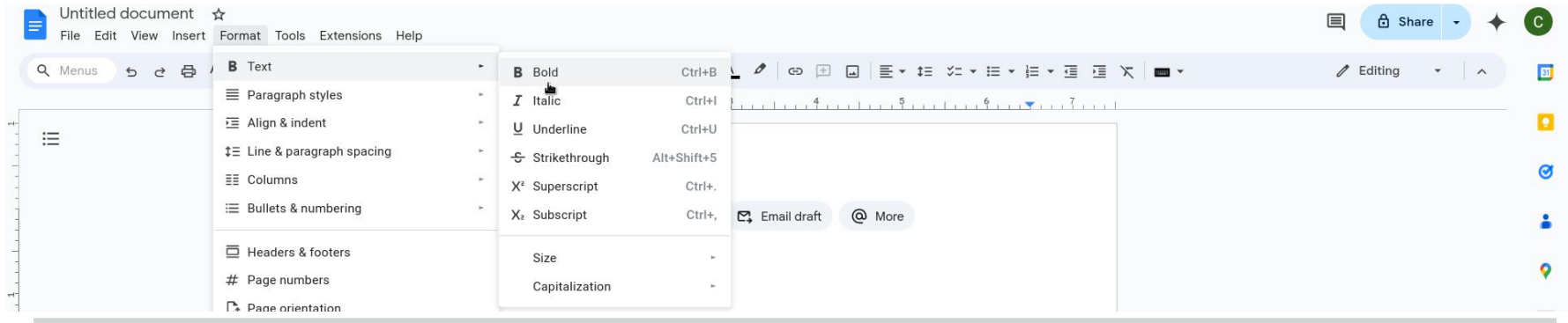


7. Flexibility and Efficiency of Use

Shortcuts — hidden from novice users — may speed up the interaction for the expert user so that the design can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

Make text bold:

- **Ctrl-B**



8. Aesthetic and Minimalist Design

Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information in an interface competes with the relevant units of information and diminishes their relative visibility.

[About](#) [Store](#)

[Gmail](#) [Images](#) [C](#)



[Google Search](#) [I'm Feeling Lucky](#)



Hurricane Milton tracker: Latest path and evacuation orders for monster storm

Milton has the potential to be one of the most destructive hurricanes to hit the Tampa area, the National Hurricane Center warned.

51 counties under a state of emergency »



Prime Day 2024: Amazon just dropped a ton of rare deals



Qantas apologizes after 'explicit' movie shown to whole flight



Humanity is approaching the upper limit of its life expectancy, study suggests



NFL power rankings: Which team has the No. 1 spot after Week 5?



Snoopy fans revolt after fan account makes pro-Trump post

Stories for you



Robert Saleh fired: Social media reacts to Jets decision to dismiss coach

The New York Jets moved on from Robert Saleh just five weeks into the 2024 NFL season. Social media was enflamed about the decision.

<https://www.yahoo.com/news/hurricane-milton-tracker-category-4-storm-heads-toward-florida-gulf-coast-massive-evacuations-ordered-160120934.html>

Trending Now

1. Hurricane Milton
2. Robert Saleh
3. Donald Trump
4. Presidential Polls
5. Sean 'Diddy' Combs
6. Bill Belichick
7. Luis Tiant
8. Lisa Marie Presley
9. MLB Playoffs
10. Howard Stern K...

Weather

Blacksburg

Today	Wed	Thu	Fri
70° 42°	71° 42°	69° 39°	72° 43°

9. Help Users Recognize, Diagnose, and Recover from Errors

Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.

The screenshot shows the Amazon checkout page for a single item. The steps are: 1. Shipping address, 2. Choose gift options, 3. Payment method, 4. Offers, and 5. Items and shipping. In step 2, the item is 'Amazon Basics Photo Picture Frame, Black, 8" x 10" Inch (Pack of 2)'. There is a 'Gift message' section with a text input field and a 'From' field. A red error message states: 'Sender name should not be empty'. To the right, there is an 'Add ons' section with a checkbox for 'Wrap in a gift bag (100% recycled) - \$4.49'. At the bottom of the gift options section is a yellow button that says 'Save gift options and continue'. On the right side of the checkout page, there is an 'Order Summary' table showing the item price, shipping & handling, total before tax, estimated tax, and the final order total of \$21.08.

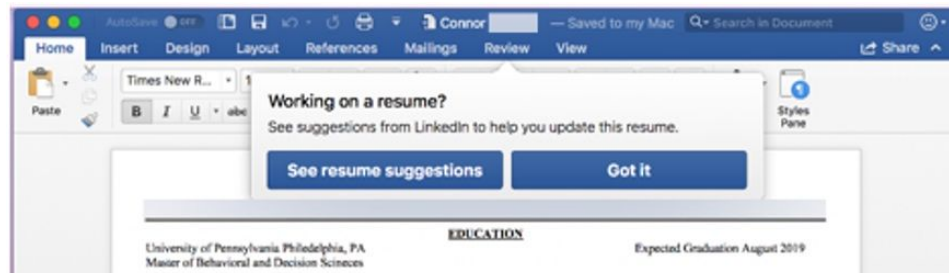
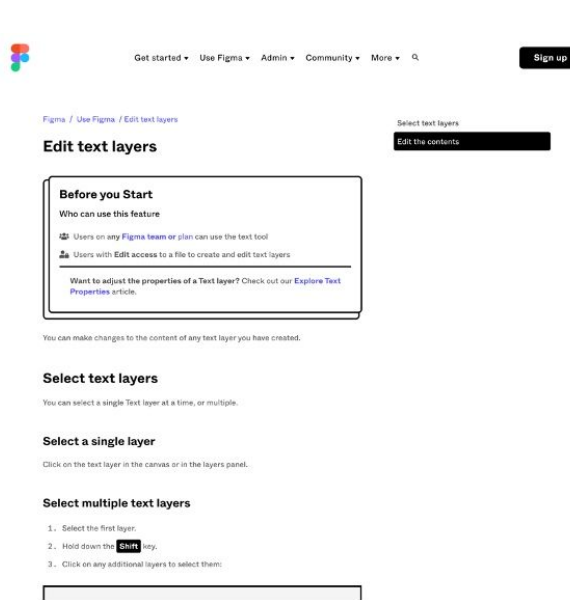
When system dialog prompts, click **Open Zoom Meetings**.



10. Help and Documentation

It's best if the system doesn't need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

- Proactive (i.e., tutorials, documentation) and reactive (i.e., FAQ, troubleshooting) help

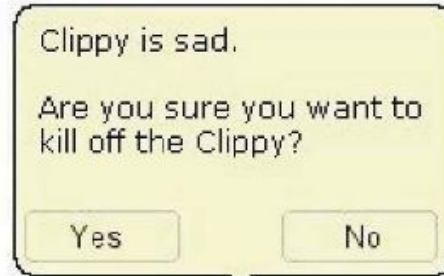
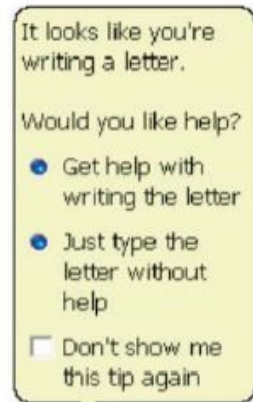


The User Interface is Important

- UIs strongly affect the perception of software
 - Usable software sells better
 - Unusable software is abandoned
- Costly to get the user interface wrong
 - Users' time is not cheap
 - Design it correctly now, or pay for it later

But User Interface Design is Hard...

UI Hall of Fame or Shame?

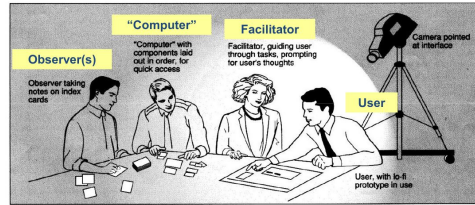


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Class Activity

Work in a your Project 1 group to design a UI for your project:

1. Use paper prototyping to brainstorm ideas for how at least one of the features of your system would look.



2. **[If AI]** Use an AI tool to generate a UI design (prototype, wireframe, or storyboard). Provide the prompt(s) provided and which AI tool.
[If !AI] Use an online tool to generate a digital design (prototype, wireframe, or storyboard) or refine your paper prototype.
3. Swap with another group/classmate to get feedback on the UI design (in particular based on the ten usability heuristics).
4. Submit the following on Canvas:
 - a. Digital UI design(s) and summary of feedback
 - b. Turn in paper prototype in class

Due: before the end of class today (submit whatever you complete)

Next Time...

- Project Workday
- **PM1.2.1 due Thursday (9/24) at 9:30am**
 - Lightning Talks during this class!!!
- **Midterm Exam (10/1) on Canvas**

References

- RS Pressman. *“Software engineering: a practitioner's approach”*.
- Cast Software *“What is Software Architecture?”*.
<<https://www.castsoftware.com/glossary/what-is-software-architecture-tools-design-definition-explanation-best>>
- K.D. Cooper, L. Torczon, *“Engineering a Compiler”*. Theo Mandel. *“Golden Rules of User Interface Design”*.
<<https://theomandel.com/resources/golden-rules-of-user-interface-design/>>
- <<https://medium.com/swlh/ordering-food-and-the-mvc-architecture-d5cbf3859d60>>
- Na Meng and Barbara Ryder
- Chris Parnin
- Sarah Heckman